SANILA WIJESEKARA

Computer Science Undergraduate

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EDUCATION

UNIVERSITY OF WESTMINSTER, UK | IIT SRI LANKA

2024 - 2028

- BSC (Hons) in Computer Science (Reading)
  - 2nd year modules
    - Machine Learning and Data Mining
    - Advanced Client Side development
    - Client Server Architectures
    - Object Oriented Programming
    - Database Systems
    - Software Development Group Project
  - 1st year modules
    - Software Development I
    - Software Development II
    - Mathematics for Computing
    - Trends in Computer Science
    - Computer System Fundamentals
    - Web design and Development

ROYAL COLLEGE, COLOMBO 7

2010-2024

- GCE A/L Examination - Maths Stream
  - Combined Mathematics - C
  - Physics - C
  - Chemistry - B
- GCE O/L Examination - 9As

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, HTML, CSS, JavaScript, SQL
- **Web & API Frameworks:** ReactJS, FastAPI
- **AI & Machine Learning libraries:** Pandas, Numpy, Scikit-learn
- **Deep Learning and Neural Networks:** Tensorflow
- **Data visualization and Annotation:** Matplotlib, Seaborn, Roboflow
- **Agentic AI Frameworks:** LangGraph, CrewAI
- **Database Management Systems:** PostgreSQL, MySQL, Superbase
- **API Testing:** Postman
- **Deployment:** Vercel, Github pages

PROFILE SUMMARY

Innovative Computer Science undergraduate with a proven track record in developing production-ready AI/ML systems and full-stack applications. Specialized in building intelligent solutions in the fields of Computer Vision, MLOps, supply chain, and Agentic AI, by leveraging the power of LangGraph. Proven performer in hackathons and competitive programming, including winning the Cipher Hackathon and placing in multiple national-level AI/ML competitions. As a Toastmaster and a proactive collaborator, I bring strong communication and public speaking skills to transform technical projects into real-world solutions.

WORK EXPERIENCE

Intern AI Engineer MAY 2026 - PRESENT

OCTAVE John Keells Group

- **AI Concierge (Cinnamon hotels):** A GenAI powered chatbot built using a **multi-agent system for cinnamon hotels across Sri Lanka and Maldives**. Involved in jailbreak testing and testing edge cases in whatsapp 360sms. Involved in bug fixing related to whatsapp answer formatting and markdown rendering.
- **MAS SPI (Stiches Per Inch) Evaluation:** A **Computer Vision** based project which involves detecting stiches real time from raw textile images taken from **MAS initimates**. Involved in building a kernel density based **clustering model** to detect multiple stich lines. Involved in improving **YOLO model performance** and fine tuning the model for different stich patterns.
- **ENERGY Consumption Usecase:** Conducted **Exploratory Data Analysis (EDA)** on energy consumption data from Cinnamon Grand hotel to identify patterns and insights for **optimizing chiller temperature setpoints,** with the objective of improving energy efficiency and reducing overall power consumption.
- **GIT tracking dashboard:** Developed an **ETL pipeline** for tracking GIT commit history, number of databricks notebooks and source files, total lines of code in scripts/notebooks and total number of pull requests.
- **Onboarding Environment:** Developed an onboarding environment to help OCTAVE technical staff gain foundational knowledge of **PySpark and Databricks,** along with guidance on setting up and configuring **Visual Studio Code** for development.
- **Techologies involved:**
  - LangGraph
  - Azure Databricks
  - Pyspark
  - Azure Devops
  - Microsoft Azure
  - YOLO Object Detection
  - Python

SOFT SKILLS

- Public Speaking
- Communication
- Leadership
- Teamwork
- Project Management
- Adaptability

CERTIFICATIONS

- **DeepLearning.AI:**
  - Machine Learning Specialization
  - Deep Learning Specialization
- **WSO2 Certified Identity Server Practitioner - V7**
- **WSO2 Certified API Manager Practitioner - V4**
- **Databricks Fundamentals**
- **AWS Educate Introduction to Generative AI**
- **Oracle Cloud Infrastructure 2024 Certified AI Foundations Associate**
- **Docker Foundations Professional Certificate**
- **Professional Certificate in Python Programming - IIT Sri Lanka**

HONOURS AND AWARDS

- **Cutting Edge 2026 - Gold Medalist**
  - Best Software Development Group Project in Computer Science Category
- **IIT Excellence Awards 2025**
  - Champion of IET Cypher Hackathon
- **Finalist (Top 10) Data Storm 6.0**
  - OCTAVE John Keells Group
- **Finalist (Top 10) Intellihack 5.0**
  - IEEE Computer Society of UCSC
- **IEEE Xtreme 18.0**
  - University rank: 54

VOLUNTEER

- **Model X:** Program Committee member
  - IEEE Computational Intelligence Society of IIT
- **HackSphere'25:** Secretarial Vice Chairperson
  - IEEE Student Branch of IIT
- **Coderally 6.0:** Organizing committee member
  - IEEE Computer Society of IIT
- **MicroMaze 2.0:** Organizing committee member
  - IEEE Robotics and Automation Society of IIT
- **Toastmaster and Speechcraft Coordinator**
  - Smedley toastmasters club
- **Medium Writer**
- **Kaggle Contributor**

REFEREES

Sankeetha Elancheliyan
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PROJECTS

Neuro Leaf Github link DECEMBER 2025 - PRESENT

Tomato leaf disease consultant

- Developed a real-time tomato leaf disease detector which detects diseases across 7 tomato disease classes using **YOLO11 Large**
- Uses a **LangGraph-based multi-agent consultation** system with intelligent routing and web fallback
- Research-driven question answering using **Retrieval Augmented Generation** (**RAG**) with FAISS as the vector database.
- **Deployed Link** - https://neuroleaf.streamlit.app/
- **Technologies:** Python, Tensorflow, CNN, YOLO, Computer Vision, RAG, LangGraph, HuggingFace, FastAPI, Streamlit, Docker

Rice Vision Github link OCTOBER 2025 - PRESENT

Satellite Based Paddy Crop Monitoring System

- RiceVision is a satellite-based paddy monitoring and yield prediction platform built to transform data-driven agriculture in Sri Lanka.
- Developed a **Long-Short Term memory** (**LSTM**) model for stage and health prediction of paddy using Sentinel-2 band datasets(B1-B12) across multiple districts in Sri Lanka.
- **Deployed link** - https://www.ricevisionlanka.com/
- **Technologies:** Python, Scikit-learn, Pandas, Numpy, Matplotlib, TensorFlow, LSTM, RNN, Superbase, PostgreSQL, React.js, TailwindCSS

Obsergent Github link OCTOBER 2026 - PRESENT

Open-Source observability tool for AI Agents and MCPs

- Developed Obsergent, an AI agent and MCP observability tool built using OpenTelemetry to trace agent execution end-to-end, capturing tool calls, inputs/outputs, execution flow, latency, and token usage and cost patterns.
- Integrated **WSO2 API manager MCP Gateway** to observe and analyse MCP tool interactions, enabling visibility into agent-to-tool communication and supporting the investigation of unexpected agent behaviour.
- Implemented agent behaviour analysis to detect tool-call loops, prompt-injection patterns, and tool-description drift, helping identify reliability and security risks in agentic workflows.
- **Technologies**: Python, OpenTelemetry, WSO2 MCP Gateway, Model Context Protocol (MCP), Agentic AI, LLM Observability, Distributed Tracing

RESEARCH AND PUBLICATIONS

An Integrated Multi-Head BiLSTM Architecture for Paddy Health monitoring, Stage Classification and Pest Outbreak Detection

ADScAI 2026 Conference Link

University of Moratuwa

- A unified **Multi-Head BiLSTM** deep learning framework that integrates Sentinel-2 optical band data, CHIRPS rainfall, and ERA5 temperature data to simultaneously monitor paddy health, classify growth stages, detect pest anomalies, and predict yield for scalable, early-warning precision agriculture in Sri Lanka.
- This research has been **accepted** for the **Poster Presentation Round at ADScAI 2026 conference,** organized by the University of Moratuwa.